

Part 1: The Five-Number Summary

In the table below, you will find the five-number summary values for the visitor counts to the 10 Carnival attractions that your class identified during the human number line activity.

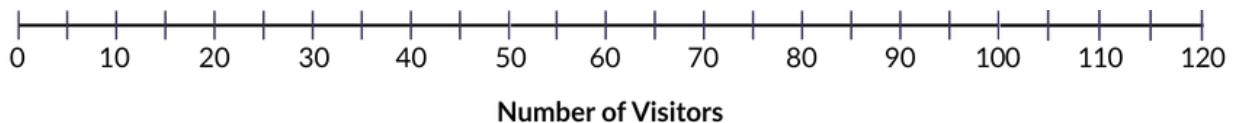
Minimum	Q1	Median	Q3	Maximum
5	78	99.5	103	114

Calculate the IQR: $IQR = Q3 - Q1 = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Interpret the IQR: The middle 50% of Carnival attractions had a spread of visitors, with counts ranging from to visitors.

Part 2: Build Your Boxplot

Using the steps below, follow along as your teacher guides you through how to draw a boxplot of the Carnival attractions data.



Step	Task	What to Do
1	Mark the Anchor Points	Place a dot above the number line at each of the values from the five-number summary.
2	Draw the Box	Draw a rectangular box using Q1 as the left edge and Q3 as the right edge. This box is the visual representation of the IQR.
3	Draw the Median Line	Draw a vertical line inside the box at the median value. This divides the box into the lower-middle 25% and the upper-middle 25%.
4	Draw the Whiskers	Draw a horizontal line from the left edge of the box (Q1) to the minimum. Draw a horizontal line from the right edge of the box (Q3) to the maximum.

Part 3: Interpret Your Boxplot

Answer the questions below.

1. Which side of the boxplot has a longer tail? In other words, which whisker is longer – the left whisker or right whisker? Calculate the following lengths to support your answer.

Left whisker length = $Q1 - \text{Minimum} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Right whisker length = Maximum - Q3 = _____ - _____ = _____

The longer whisker is on the **LEFT SIDE** / **RIGHT SIDE** of the boxplot. (Circle one.)

2. What does the length of the longer side/whisker tell us about the **SHAPE** of this distribution?

SKEWED LEFT

SYMMETRIC

SKEWED RIGHT

Explain your reasoning: _____

3. Where does the median line sit inside the box? Calculate the following lengths to support your answer.

Left side of box length = Median - Q1 = _____ - _____ = _____

Right side of box length = $Q3 - \text{Median} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

The median is **CLOSER** to Q1 (left edge) / **ABOUT** in the **MIDDLE** / **CLOSER** to Q3 (right edge).

4. What does the placement of the median line tell us about the **SHAPE** of this distribution?

SKEWED LEFT

SYMMETRIC

SKEWED RIGHT

Explain your reasoning: _____
